Formula 1 proj database

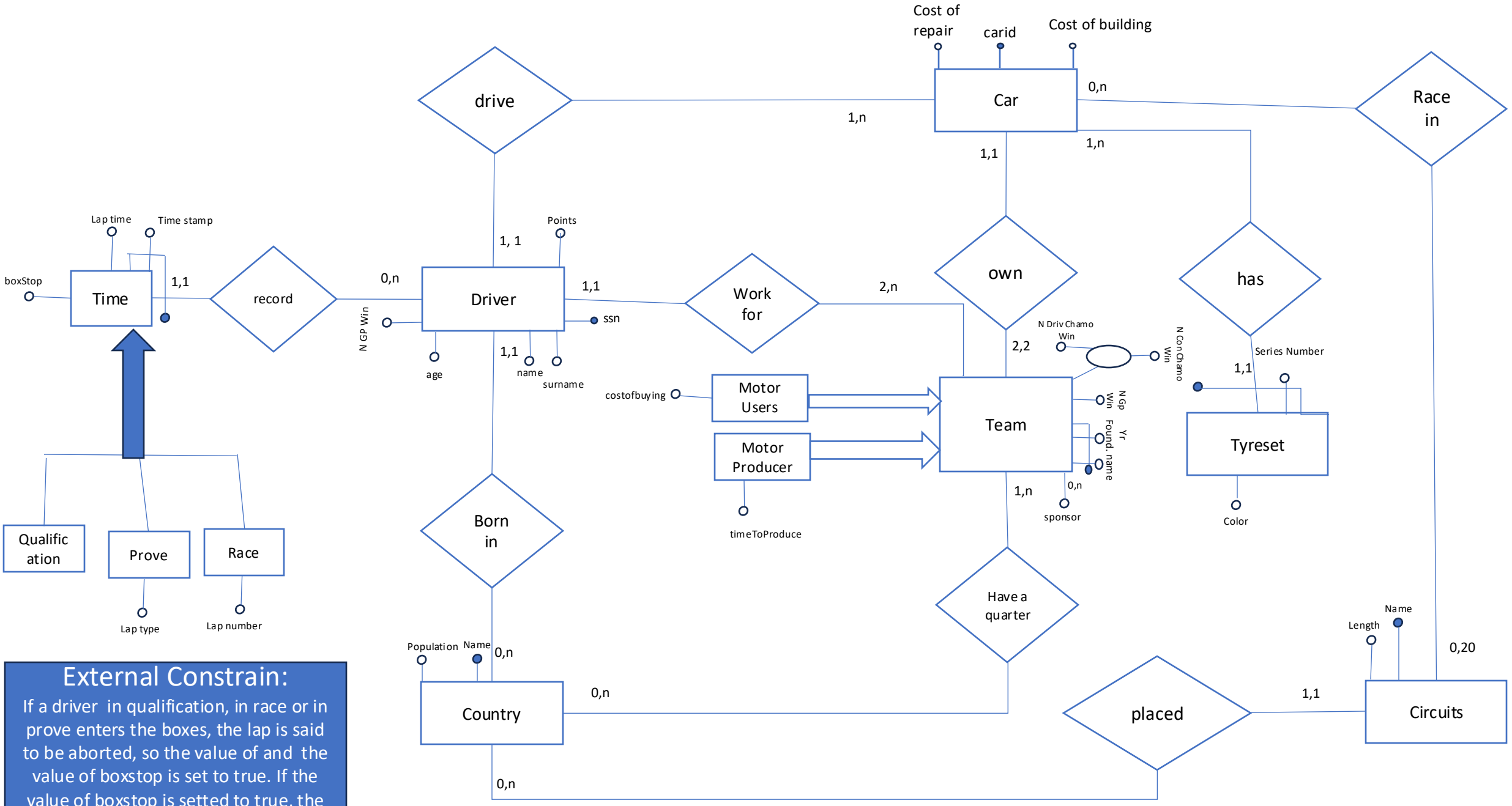
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First phase:
Conceptual Design

Text

The conceptual schema represents an application dealing with Formula 1 data. We aim to model a database to track various aspects of Formula One during a single season. The database is valid for one season only and is reset for the next season. The schema follows this specification:

- We are interested in Cars, which are owned by a Team. Each team has a name, year of foundation, number of Grand Prix wins, number of Constructors' Championships, number of Drivers' Championships, and can have sponsors. Each team can either produce its own Engine or use one from another team.
- Each Car can be driven by one or more Drivers. Each driver has a name, surname, age, and number of Grand Prix wins. A driver can drive one and only one car.
- We are interested also in the cost of building and repair the car.
- Each driver can record multiple lap Times. For each recorded time, we are interested in the lap time, whether it occurred during a box stop (true or false), and the timestamp
- A lap time can be recorded during practice, qualification, or a race. In the case of practice laps, we are interested in the type (whether it is for qualification or race preparation). For race lap times, we need to record the specific lap during which the time was set (e.g., if a driver records a time on lap 15, the value is 15). The lap number is specified only for race laps.
- Each car can compete in multiple Circuits. For each circuit, we are interested in its name and length. A circuit is located in one and only one country.
- Each driver is born in one and only one country.
- Each team must have at least one headquarters located in a country
- Each car must have at least one Tire Set. For each tire set, we record the color and serial number. A tire set is owned by one and only one car
- For each country in the database we are also interested in his population.
- Each team can be an engine user, engine producer, or both. For engine users, we are interested in the cost of purchasing the engine. For engine producers, we record the name of the engine produce



External Constrain:

If a driver in qualification, in race or in prove enters the boxes, the lap is said to be aborted, so the value of and the value of boxstop is set to true. If the value of boxstop is setted to true, the laptime is set to NULL.

Glossary of terms

Type	Definition
Driver	A person who drives a car in Formula 1 events, associated with one and only one car.
Team	An entity that owns cars and can produce or use engines; it may also have sponsors and must have at least a headquarters in a country.
Car	A vehicle owned by a team, driven by drivers, and equipped with tire sets.
Lap Time	The recorded time for a driver completing a lap; includes additional data like box stops, type (practice, qualification, race), and timestamp.
Circuit	A race track where cars compete; each circuit is uniquely identified by its name and is located in one country.
Tire Set	A set of tires assigned to one car, identified by serial number and color.
Country	Geographic information associated with circuits, teams, or driver origins.

Data Dictionary: Entities 1

- **Driver**:
  - Attributes: Name, Surname, Age, Number of Grand Prix Wins.
  - External Constraint: Born in one and only one country.
- **Team**:
  - Attributes: Name, Year of Foundation, Number of Grand Prix Wins, Constructors' Championships, Drivers' Championships, Sponsors.
  - Relationships: Owns cars, may produce or use engines.
- **Car**:
  - Attributes: Car ID, Cost of building, Cost of repair
  - Relationships: Owned by one team, driven by one or more drivers, competes in circuits, and has tire sets.
- **Lap Time**:
  - Attributes: Lap Time, Box Stop (true/false), Timestamp, Lap Number, Lap Type.
  - External Constraint: If a lap is aborted (box stop is true), lap time is set to NULL.
- **Circuit**:
  - Attributes: Name, Length.
  - External Constraint: Located in one and only one country.
- **Tire Set**:
  - Attributes: Serial Number, Color.
  - Relationships: Owned by one and only one car.
- **Country**:
  - Attributes: Name, population
  - Relationships: Associated with drivers, circuits, or team headquarters.

Data Dictionary: Entities 2

- **Race**:
  - Attributes: Lap number
  - External Constraint: Associated with one and only one time.
  - Relationship: GEN
- **Qualification**:
  - External Constraint: Associated with one and only one time.
  - Relationship: GEN
- **Prove**:
  - Attributes: lap type
  - External Constraint: Associated with one and only one time.
  - Relationship: GEN
- **Motor-users**:
  - Attributes: cost of buying
  - Relationship: ISA
- **Motor-producer**:
  - Attributes: time to produce
  - Relationship: ISA

Data Dictionary: Relationships

- **Own**: Between Team and Car.
- **Drive**: Between Driver and Car.
- **Record**: Between Driver and Lap Time
- **Race In**: Between Car and Circuit
- **Has**: Between Car and Tire Set
- **Placed:** Between Country and Circuits
- **Born in:** Between Country and Driver
- **Have a quarter:** Between Country and Team

Data Dictionary: External Constraints

- **First:** If a driver in qualification, in race or in prove enters the boxes, the lap is said to be aborted, so the value of and the value of boxstop is set to true.
- **Second:** If the value of boxstop is setted to true, the laptime is set to NULL.

Tables of volumes 1

Name	Type	Volumes
Driver	Entity	30
Team	Entity	10
Car	Entity	20
Time	Entity	50 000
Circuit	Entity	20
Tire set	Entity	2000
Country	Entity	205
Qualification	Entity	7000
Prove	Entity	7000
Race	Entity	36000
Motor User	Entity	10
Motor Producer	Entity	10

Tables of volumes 2

Name	Type	Volumes
Own	Relationship	20
Drive	Relationship	30
Record	Relationship	50 000
Race in	Relationship	400
Has	Relationship	2000
Placed	Relationship	20
Born in	Relationship	30
Work for	Relationship	30
Have a quarter	Relationship	150

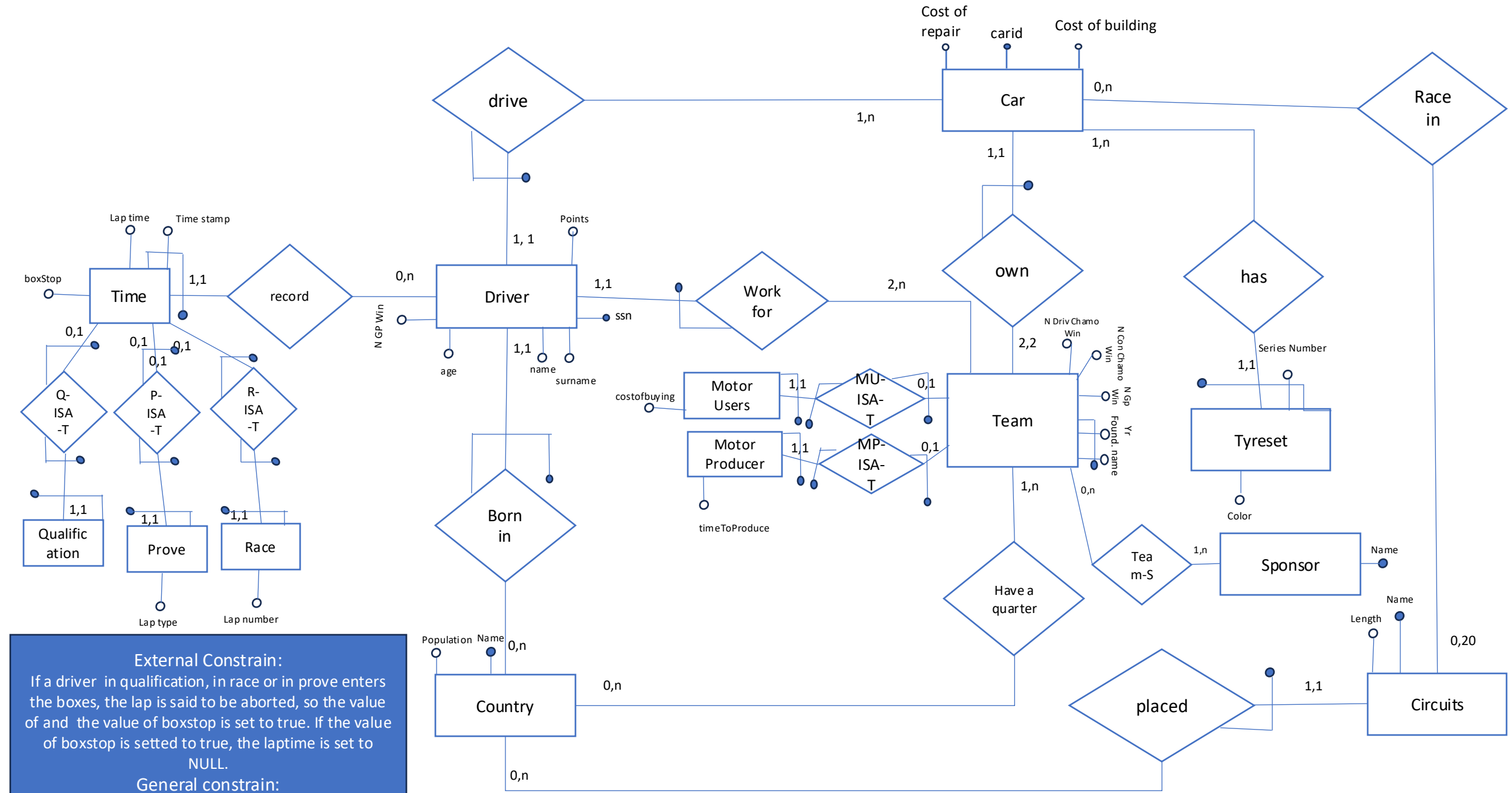
List of operations

1. Find the average lap time for each driver across all circuits where the driver has recorded at least 10 laps, excluding aborted laps.
2. List the drivers who have achieved the fastest lap in at least 3 different races during a single season, including the details of the circuits in that race (how long was the circuit).
3. Identify the circuits where a specific driver has achieved the fastest lap in both qualification and race sessions.
4. Calculate the total points scored by each team.
5. Find out the Country where the sum of all point of the drivers born there is maximum.

Table of operations

Operation	Type	Frequency	Details
1	Interactive	100/season	This operation is typically used for post-race analysis or comparing driver performance across circuits during a season. Analysts or teams may request this in real-time to understand driver consistency
2	Interactive	10/season	This query helps identify standout drivers who excel in setting the fastest laps. It is commonly used for performance analysis or media coverage highlighting exceptional achievements.
3	Interactive	5/season	This query is aimed at assessing driver dominance on specific circuits. It is often requested for detailed performance reviews, focusing on both race-day and qualification excellence
4	Batch	20/season	This operation is frequently used to monitor the constructors' championship standings throughout the season. It is often run automatically after every race to update leaderboards.
5	Batch	5/season	This query is useful to see which country produce good drivers.

Second phase:
Restructuring of the conceptual schema



External Constrain:

If a driver in qualification, in race or in prove enters the boxes, the lap is said to be aborted, so the value of and the value of boxstop is set to true. If the value of boxstop is setted to true, the laptime is set to NULL.

General constrain:

Each instance of Time partecipates to Q-ISA-T or P-ISA-T or R-ISA-T but not to more then one

Data Dictionary restructuring: Entities 1

- **Driver**:
  - Attributes: Name, Surname, Age, Number of Grand Prix Wins.
  - External Constraint: Born in one and only one country.
- **Team**:
  - Attributes: Name, Year of Foundation, Number of Grand Prix Wins, Constructors' Championships, Drivers' Championships, Sponsors.
  - Relationships: Owns cars, may produce or use engines.
- **Car**:
  - Attributes: Car ID.
  - Relationships: Owned by one team, driven by one or more drivers, competes in circuits, and has tire sets.
- **Lap Time**:
  - Attributes: Lap Time, Box Stop (true/false), Timestamp, Lap Number, Lap Type.
  - External Constraint: If a lap is aborted (box stop is true), lap time is set to NULL.
- **Circuit**:
  - Attributes: Name, Length.
  - External Constraint: Located in one and only one country.
- **Tire Set**:
  - Attributes: Serial Number, Color.
  - Relationships: Owned by one and only one car.
- **Country**:
  - Attributes: Name.
  - Relationships: Associated with drivers, circuits, or team headquarters

Data Dictionary restructuring: Entities 2

- **Race**:
  - Attributes: Lap number
  - External Constraint: Associated with one and only one time.
  - Relationship: R-ISA-T
- **Qualification**:
  - External Constraint: Associated with one and only one time.
  - Relationship: Q-ISA-T
- **Prove**:
  - Attributes: lap type
  - External Constraint: Associated with one and only one time.
  - Relationship: P-ISA-T
- **Motor-users**:
  - Attributes: cost of buying
  - Relationship: MU-ISA-T
- **Motor-producer**:
  - Attributes: timeToProduce
  - Relationship: MP-ISA-T
- **Sponsor**:
  - Attributes: name
  - Relationship: TEAM-S

Data Dictionary restructuring: Relationships

- **Own**: Between Team and Car.
- **Drive**: Between Driver and Car.
- **Record**: Between Driver and Lap Time
- **Race In**: Between Car and Circuit
- **Has**: Between Car and Tire Set
- **Placed:** Between Country and Circuits
- **Born in:** Between Country and Driver
- **Have a quarter:** Between Country and Team
- **Q-ISA-T:** Between Qualification and Time
- **P-ISA-T:** Between Prove and Time
- **R-ISA-T:** Between Race and Time
- **MU-ISA-T:** Between Motor user and Team
- **MP-ISA-T:** Between Motor producer and Team
- **TEAM-S**: Between Sponsor and Team

Data Dictionary: External and General Constraints

- **First:** If a driver in qualification, in race or in prove enters the boxes, the lap is said to be aborted, so the value of and the value of boxstop is set to true.
- **Second:** If the value of boxstop is setted to true, the laptime is set to NULL.
- **Third**: Each instance of Time partecipates to Q-ISA-T or P-ISA-T or R-ISA-T but not to more then one

Tables of volumes 1

Name	Type	Volumes
Driver	Entity	30
Team	Entity	10
Car	Entity	20
Time	Entity	50 000
Circuit	Entity	20
Tire set	Entity	2000
Country	Entity	205
Qualification	Entity	7000
Prove	Entity	7000
Race	Entity	36000
Motor User	Entity	10
Motor Producer	Entity	10
Sponsor	Entity	100

Tables of volumes 2

Name	Type	Volumes
Own	Relationship	20
Drive	Relationship	30
Record	Relationship	50 000
Race in	Relationship	400
Has	Relationship	2000
Placed	Relationship	20
Born in	Relationship	30
Work for	Relationship	30
Have a quarter	Relationship	150
R-ISA-T	Relationship	36000
Q-ISA-T	Relationship	7000
P-ISA-T	Relationship	7000
MU-ISA-T	Relationship	10
MP-ISA-T	Relationship	10
TEAM-S	Relationship	100

Access table Operation 1

1. Find the average lap time for each driver across all circuits where the driver has recorded at least 10 laps, excluding aborted laps.

Concept	Construct	Access	Type
Race	Entity	36000	R
R-ISA-T	Relationship	36000	R
Time	Entity	36000	R
Record	Relationship	36000	R
Driver	Entity	20	R
Drive	Relationship	20	R
Car	Entity	20	R
Racein	Relationship	20	R
Circuits	Entity	20	R

Access table Operation 2

2. List the drivers who have achieved the fastest lap in at least 3 different races during a single season, including the details of the circuits in that race (how long was the circuit)

Concept	Construct	Access	Type
Race	Entity	36000	R
R-ISA-T	Relationship	20	R
Time	Entity	20	R
Record	Relationship	20	R
Driver	Entity	20	R
Drive	Relationship	20	R
Car	Entity	20	R
Racein	Relationship	20	R
Circuits	Entity	20	R

Access table Operation 3

3. Identify the circuits where a specific driver has achieved the fastest lap in both qualification and race sessions.

Concept	Construct	Access	Type
Race	Entity	36000	R
R-ISA-T	Relationship	1	R
Qualification	Entity	7000	R
Q-ISA-T	Relationship	1	R
Time	Entity	1	R
Record	Relationship	1	R
Driver	Entity	1	R
Drive	Relationship	1	R
Car	Entity	1	R
Racein	Relationship	20	R
Circuits	Entity	20	R

Access table Operation 4

4. Calculate the total points scored by each team

Concept	Construct	Access	Type
Team	Entity	10	R
WorkFor	Relationship	30	R
Driver	Entity	30	R

Access table Operation 5

5. Find out the Country where the sum of all point of the drivers born there is maximum.

Concept	Construct	Access	Type
Driver	Entity	30	R
BornIn	Relationship	30	R
Country	Entity	30	R

Third phase:
Direct translation

Relational model

- Car(carid, costOfRepay, costOfBuilding)
  - FK: Car[carid] ⊆ Own[car]
  - Inclusion: Car[carid] ⊆ Drive[car]
- Circuit(name, length)
  - FK: Circuits[name] ⊆ Placed[circuit]
  - External constraint: ∀c∈Circuits,|{x∈RaceIn|x.Circuit=c}|≤20
- Country(name, population)
- Driver(ssn, points, surname, name, age, nGPWin, carid)
  - FK: Driver[ssn] ⊆ WorkFor[driver]
  - FK: Driver[ssn] ⊆ BornIn[driver]
  - FK: Driver[ssn] ⊆ Drive[driver]
- Team(yrFound, name, nGPWin,nConChampWin, nDrivChampWin)
  - Inclusion: Team[yrFound,name] ⊆ HaveaQuarter[yrFound,name]
  - External constraint: ∀t∈Team,|{c∈Own|c.Team=t}|=2
- Sponsor(name)
  - Inclusion: Sponsor[name] ⊆ Team-S[sponsor]

Relational model

- Tyreset(seriesNumber, car, color)
  - FK: Tyreset[car] ⊆ Car[carid]
- Time(timeStamp, driver, lapTime, boxStop)
  - FK: Time[driver] ⊆ Driver[ssn]
  - External constraint: (Time[lapTime] = TRUE AND Time[boxStop] IS NULL)
    OR (Time[lapTime] = FALSE AND Time[boxStop] IS NOT NULL)
- Qualification(timeStamp, driver)
  - FK: Qualification[timeStamp, driver] ⊆ Time[timeStamp, driver]
  - Generalization Constraint: Qualification[timeStamp, driver] ∩ Prove [timeStamp, driver] = 0
  - Generalization Constraint: Qualification[timeStamp, driver] ∩ Race [timeStamp, driver] = 0
- Prove(timeStamp, driver, lapType)
  - FK: Prove[timeStamp, driver] ⊆ Time[timeStamp, driver]
  - Generalization Constraint: Prove[timeStamp, driver] ∩ Race [timeStamp, driver] = 0
- Race(timeStamp, driver, lapNumber)
  - FK: Race[timeStamp, driver] ⊆ Time[timeStamp, driver]

Relational model

- MotorUser(nameT, yearFoundT, costOfBuying)
  - FK: MotorUser[nameT, yearFoundT] ⊆ Team[name, yearFound]
- MotorProducer(yearFoundT, nameT, timeToProduce)
  - FK: MotorProducer[nameT, yearFoundT] ⊆ Team[name, yearFound]
- RaceIn(carId, nameCirc)
  - FK: RaceIn[carId]⊆ Car[carId]
  - FK: RaceIn[nameCirc]⊆ Circuit[name]
- HaveAQuarter(teamN, yrFound, country)
  - FK: HaveAQuarter[teamN, yrFound]⊆ Team[name,yrFound]
  - FK: HaveAQuarter[country]⊆ Country[name]
- Team-S(teamN, teamYrFound, sponsor)
  - FK: Team-S[teamN, teamYrFound] ⊆ Team[name, yearFound]
  - FK: Team-S[sponsor] ⊆ Sponsor[name]

Relational model

- BornIn(driver, country)
  - FK: BornIn[driver]⊆ Driver[ssn]
  - FK: BornIn[country] ⊆ Country[name]
- WorkFor(driver, teamN, teamYrFound)
  - FK: WorkFor[driver]⊆ Driver[ssn]
  - FK: WorkFor[teamN, teamYrFound] ⊆ Team[name, yearFound]
- Own(car, teamN, teamYrFound)
  - FK: Own[car]⊆ Car[carId]
  - FK: Own[teamN, teamYrFound] ⊆ Team[name, yearFound]
- Placed(circuit, countryName)
  - FK: Placed[circuit]⊆ Circuit[name]
  - FK: Placed[country]⊆ Country[name]
- Drive(driver, car)
  - FK: Drive[driver]⊆ Driver[ssn]
  - FK: Drive[car]⊆ Car[carid]

Specification Application Load Operation 1

1. Find the average lap time for each driver across all circuits where the driver has recorded at least 10 laps, excluding aborted laps.

Concept	Construct	Access	Type
Drive	Relationship	20	R
RaceIn	Relationship	20	R
Race	Entity	36000	R

Specification Application Load Operation 2

2. List the drivers who have achieved the fastest lap in at least 3 different races during a single season, including the details of the circuits in that race (how long was the circuit)

Concept	Construct	Access	Type
Drive	Relationship	20	R
RaceIn	Relationship	20	R
Race	Entity	36000	R
Circuit	Entity	20	R

Specification Application Load Operation 3

3. Identify the circuits where a specific driver has achieved the fastest lap in both qualification and race sessions.

Concept	Construct	Access	Type
Drive	Relationship	20	R
RaceIn	Relationship	20	R
Race	Entity	36000	R
Qualification	Entity	7000	R

Specification Application Load Operation 4

4. Calculate the total points scored by each team

Concept	Construct	Access	Type
WorkFor	Relationship	30	R
Driver	Entity	30	R

Specification Application Load Operation 5

5. Find out the Country where the sum of all point of the drivers born there is maximum.

Concept	Construct	Access	Type
Driver	Entity	30	R
BornIn	Relationship	30	R

Fourth phase:
Restructuring of the relational schema

Restructuring 1,2,3 operation

- It is useful to put together a new table with driver, carId, circuits, because in the first three operations, we use them together and so it cost less to have just one table with all three the data then to have to access two table.
- RaceParticipation(driver, carid, circuit)
  - FK: RaceParticipation[driver]⊆ Driver[ssn]
  - FK: RaceParticipation[car]⊆ Car[carid]
  - FK: RaceParticipation[circuit] ⊆ Circuit[name]

Restructuring 5 Operation

- In driver add the country so that we reduce the join in the fifth operation.
- Driver(ssn, points, surname, name, age, nGPWin, carid, country)
  - FK: Driver[ssn] ⊆ WorkFor[driver]
  - FK: Driver[ssn] ⊆ BornIn[driver]
  - FK: Driver[ssn] ⊆ Drive[driver]
  - FK: Driver[country] ⊆ Country[name]

Restructuring of relations

- I have also restructured three operation in the entity table so that we don't have more table then what we need.
- For Team -> have a quarter
  - Team(yrFound, name, nGPWin,nConChampWin, nDrivChampWin, country)
    - FK: Team[country] ⊆ Country[name]
    - External constraint: ∀t∈Team,|{c∈Own|c.Team=t}|=2
- For Car -> own
  - Car(carid, costOfRepay, costOfBuilding, teamName, teamYr)
    - FK: Car[teamName, teamYr] ⊆ Team[name, yrFound]
    - Inclusion: Car[carid] ⊆ Driver[carid]
- For Circuit -> placed
  - Circuit(name, length, country)
    - FK: Circuit[country] ⊆ Country[name]
    - External constraint: ∀c∈Circuits,|{x∈RaceIn|x.Circuit=c}|≤20
- For Driver -> drive
  - Driver(ssn, points, surname, name, age, nGPWin, carid, country)
    - FK: Driver[ssn] ⊆ WorkFor[driver]
    - FK: Driver[ssn] ⊆ BornIn[driver]
    - FK: Driver[carid] ⊆ Car[carid]
    - FK: Driver[country] ⊆ Country[name]

Relational model restructured

- Car(carid, costOfRepay, costOfBuilding, teamName, teamYr)
  - FK: Car[teamName, teamYr] ⊆ Team[name, yrFound]
  - Inclusion: Car[carid] ⊆ Driver[carid]
- Circuit(name, length, country)
  - FK: Circuit[country] ⊆ Country[name]
  - External constraint: ∀c∈Circuits,|{x∈RaceIn|x.Circuit=c}|≤20
- Country(name, population)
- Driver(ssn, points, surname, name, age, nGPWin, carid, country)
  - FK: Driver[ssn] ⊆ WorkFor[driver]
  - FK: Driver[carid] ⊆ Car[carid]
  - FK: Driver[country] ⊆ Country[name]
- Team(yrFound, name, nGPWin, nConChampWin, nDrivChampWin, country)
  - FK: Team[country] ⊆ Country[name]
  - External constraint: ∀t∈Team,|{c∈Car|c.Team=t}|=2
- Sponsor(name)
  - Inclusion: Sponsor[name] ⊆ Team-S[sponsor]

Relational model restructured

- Tyreset(seriesNumber, car, color)
  - FK: Tyreset[car] ⊆ Car[id]
- Time(timeStamp, driver, lapTime, boxStop)
  - FK: Time[Driver]⊆ Driver[SSN]
  - External constraint: (Time[lapTime]= TRUE AND Time[boxStop] IS NULL OR Time[lapTime]= TRUE AND Time[boxStop] IS NULL
- Qualification(timeStamp, driver)
  - FK: Qualification[timeStamp, driver] ⊆ Time[timeStamp, driver]
  - Generalization Constraint: Qualification[timeStamp, driver] ∩ Prove[timeStamp, driver]=0
  - Generalization Constraint: Qualification[timeStamp, driver] ∩ Race[timeStamp, driver]=0
- Prove(timeStamp, driver, lapType)
  - FK: Prove[timeStamp, driver] ⊆ Time[timeStamp, driver]
  - Generalization Constraint: Prove[timeStamp, driver] ∩ Race[timeStamp, driver]=0
- Race(timeStamp, driver, lapNumber)
  - FK: Prove[timeStamp, driver] ⊆ Time[timeStamp, driver]

Relational model restructured

- MotorUser(nameT, yearFoundT, costOfBuying)
  - FK: MotorUser[nameT, yearFoundT] ⊆ Team[name, yearFound]
- MotorProducer(yearFoundT, nameT, timeToProduce)
  - FK: MotorProducer[nameT, yearFoundT] ⊆ Team[name, yearFound]
- RaceIn(carId, nameCirc)
  - FK: RaceIn[carId]⊆ Car[carId]
  - FK: RaceIn[nameCirc]⊆ Circuit[name]
- Team-S(teamN, teamYrFound, sponsor)
  - FK: Team-S[teamN, teamYrFound] ⊆ Team[name, yearFound]
  - FK: Team-S[sponsor] ⊆ Sponsor[name]

Relational model restructured

- WorkFor(driver, teamN, teamYrFound)
  - FK: WorkFor[driver]⊆ Driver[ssn]
  - FK: WorkFor[teamN, teamYrFound] ⊆ Team[name, yearFound]
- RaceParticipation(driver, carid, circuit)
  - FK: RaceParticipation[driver]⊆ Driver[ssn]
  - FK: RaceParticipation[car]⊆ Car[carid]
  - FK: RaceParticipation[circuit] ⊆ Circuit[name]